



Available online at
 ScienceDirect
 www.sciencedirect.com

Elsevier Masson France
 EM|consulte
 www.em-consulte.com



Original article

Reversal of multidrug resistance by small interfering RNA (siRNA) in doxorubicin-resistant MCF-7 breast cancer cells

Yaprak Dönmez, Ufuk Gündüz *

Middle East Technical University, Department of Biological Sciences, 06531 Ankara, Turkey

ARTICLE INFO

Article history:

Received 19 July 2010

Accepted 6 December 2010

Available online 30 December 2010

Keywords:

MDR reversal
 P-glycoprotein
 siRNA

ABSTRACT

Purpose: Resistance to anticancer drugs is a serious obstacle to cancer chemotherapy. A common form of multidrug resistance (MDR) is caused by the overexpression of transmembrane transporter proteins P-glycoprotein (P-gp) and multidrug resistance-associated protein-1 (MRP1), encoded by *MDR1* and *MRP1* genes, respectively. These proteins lead to reduced intracellular drug concentration and decreased cytotoxicity by means of their ability to pump the drugs out of the cells. Breast cancer tumor resistance is mainly associated with overexpression of P-gp/MDR1. Although some chemical MDR modulators aim to overcome MDR by interfering functioning of P-gp, their toxicities limit their usage in clinics. Consequently, RNA interference mediated sequence specific inhibition of the expression of P-gp/MDR1 mRNA may be an efficient tool to reverse MDR phenotype and increase the success of chemotherapy. Aim of this study was resensitizing doxorubicin-resistant breast cancer cells to anticancer agent doxorubicin by selective downregulation of P-gp/MDR1 mRNA.

Methods: The effect of the selected MDR1 siRNA, and MRP1 expression after MDR1 silencing was determined by qPCR analysis. Intracellular drug accumulation and localization was investigated by confocal laser scanning microscopy after treatment with MDR1 siRNA. XTT cell proliferation assay was performed to determine the effect of MDR1 silencing on doxorubicin sensitivity.

Results: The results demonstrated that approximately 90% gene silencing occurred by the selected siRNA targeting MDR1 mRNA. However, the level of MRP1 mRNA did not change after MDR1 downregulation. Silencing of P-gp encoding *MDR1* gene resulted in almost complete restoration of the intracellular doxorubicin accumulation and relocation of the drug in the nuclei. Introduction of siRNA resulted in about 70% resensitization to doxorubicin.

Conclusions: Selected siRNA duplex was shown to effectively inhibit *MDR1* gene expression, restore doxorubicin accumulation and localization, and enhance chemosensitivity of resistant cells, which makes it a suitable candidate for therapeutic applications.

© 2010 Elsevier Masson SAS. All rights reserved.

1. Introduction

Drug resistance is a significant factor that limits the effectiveness of chemotherapeutic drugs. Tumors may be intrinsically resistant to chemotherapy prior to treatment, or drug resistance can be induced by chemotherapeutic drug during treatment, so that tumors that are initially sensitive become resistant to chemotherapy [1]. Currently, drug resistance to chemotherapy is believed to cause failure in over 90% of patients with metastatic cancers [1,2]. Simultaneous resistance of drug resistant tumors to a number of structurally and functionally unrelated chemotherapeutic agents is named as multidrug resistance (MDR) [3]. Mechanisms of drug resistance include ATP driven transporter proteins pumping drugs out of the cells, increased trapping of

drugs in intracellular organelles away from the cytosol and nucleoplasm, alkaline shift of cellular pH that reduces the intracellular drug accumulation, enhancement of detoxification systems, activation of DNA repair mechanisms, altered apoptosis regulation and changes in plasma membrane structure which affect drug permeability [4,5]. Transport-based MDR is generally associated with overexpression of efflux proteins belonging to the ATP binding cassette (ABC) family of transporters [6]. The major efflux pump in MDR is 170 kDa P-glycoprotein (P-gp), encoded by *ABCB1* (*MDR1*) gene. The broad substrate spectrum of Pgp comprises anthracyclines (doxorubicin, daunorubicin, epirubicin), taxenes (paclitaxel, docetaxel), *Vinca* alkaloids (vincristine, vinblastine, etc.), epipodophyllotoxins (etoposide, teniposide) and mitomycin C [7]. P-gp is expressed in a wide variety of normal tissues including intestine, liver, kidney, placenta and blood-brain barrier as well as in numerous tumors [3]. In nearly 40% of breast cancer tumors, P-gp is overexpressed [8,9]. Also, overexpression of 190-kDa MRP1, encoded by *ABCC1* (*MRP1*) gene, and its role in

* Corresponding author. Tel.: +90 3122 105 183; fax: +90 3122 107 976.

E-mail address: ufukg@metu.edu.tr (U. Gündüz).